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Original article

Beyond LCOE: A study of ocean energy technology development and deployment attractiveness

Adrian de Andres [a,*], Andy MacGillivray [b], Owain Roberts [b], Raul Guanche [a], Henry Jeffrey [b]

[a] Environmental Hydraulics Institute - IH Cantabria, University de Cantabria, Santander, Spain
[b] Institute of Energy Systems, University of Edinburgh, Edinburgh, UK

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ABSTRACT

Ocean (wave and tidal) energy is still in the prototype testing stage, with several challenges standing between the sector's current status and the goal of commercial deployment. The study of how to reduce the cost of energy, to a level competitive with renewable energy alternatives, is an important aspect of sector development. A comprehensive PESTLE (Political, Economic, Social, Technological, Legal and Environmental) analysis has been carried out to reveal the risks and uncertainties that face wave and tidal energy technology within identified themes. The purpose of the PESTLE analysis was to ascertain the factors most relevant to the reduction of the Levelised Cost of Energy (LCOE), and the wider commercialisation of wave and tidal energy technology. It has been shown that two separate case studies are required in order to more accurately assess the needs of the ocean energy sector. Firstly the suitability of different geographic locations for technology deployment and secondly the progress of the technology in the development cycle; the areas that require the focus of research and development activity. With the completion of both case studies, the paths towards sector development and cost effective energy production has been presented.

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Introduction

Ocean energy is still in a prototype testing stage and there has been little commercial deployment to date. One of the main reasons for this is that, currently, the cost of energy produced by wind and tidal energy devices is not cost-competitive with other alternative renewable energy sources such as wind. However, there are very few scientific studies regarding the current status of the Levelised Cost of Energy (LCOE) and its evolution for present day ocean energy technologies.

Research carried out by Carbon Trust concluded that ocean energy could achieve an equivalent cost of 19 €c/kW h by 2050 [1]. This cost is still high when compared to the cost of existing coal power generation (6–8 cents/kW h), gas power generation (8–10 €c/kW h), wind (5–11 €c/kW h) and solar PV (8–12 cents/kW h) [2] - but is nevertheless a significant improvement on the current cost of energy of early ocean energy prototypes. There is a global deployment potential for ocean energy of 337 GW [3], which could result in the generation of over 885 TW h/year. Taking this into consideration, it is clear that there is a need for research that outlines the pathway required to achieve LCOE and deployment goals.

An example of a study which sets out the path for achieving commercialisation is the Energy Technologies Institute (ETI) Roadmap [4], which presents 40 technology and deployment issues and prioritises them from the perspective of the ETI. While this reflects the needs of the industry, there is no route map for defining the timelines for the development of each identified issue, or prioritisation of the aspects that are considered to be a key step towards achieving a lower LCOE. To accomplish this, funding should be targeted on the areas where the impact on reducing the LCOE is highest.

A study by Previsic and Shoele [3], which examines the current status of wave energy LCOE, identified certain key with cost-reduction potential, but for a specific type of converter. It was concluded that wave energy could achieve a cost of energy equivalent to 12 €c/kW h in the near future [3]. Similarly, Li et al. [4] studied the economic ability of a tidal farm. The study found that the minimum cost of energy for a large farm (consisting of 100 turbines) with a 20-year life in offshore British Columbia is about 8 cents/kW h, which is only slightly higher than the local market price. The study highlighted that the two most important variables

* Corresponding author.
E-mail addresses: adrian.deandres@ed.ac.uk (A. de Andres), A.MacGillivray@ed.ac.uk (A. MacGillivray), owain.roberts@ed.ac.uk (O. Roberts), guancher@unican.es (R. Guanche), henry.jeffrey@ed.ac.uk (H. Jeffrey).

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affecting cost of energy are farm size and the relative distance between devices.

The economic study by Guanche et al. [5] investigated the influence of met-ocean variability and feed in tariff on LCOE. It was concluded that, for the prototype analysed within the study, a feed-in-tariff of 0.45 €c/kW h would be needed to return a profit, which is significantly higher than any feed-in-tariff available at the present time. This demonstrates that, for the technology in question, commercialisation remains remote and highlights the need for continued government support to enable the first deployments and facilitate future cost reductions.

As well as government support, investors in ocean energy technologies are needed in order to assist the early deployments of devices and arrays. However, there is significant risk to investing in ocean energy, not only due to the substantial capital required, but also the unpredictability of the costs and future revenue streams [6]. This creates something of a paradox, investors in early stage developers are in greatest need of future cost projections in order to have reassurance, but this is the stage at which future projects are their most uncertain. Therefore the systematic identification of the risks and uncertainties and an understanding on their potential impacts on overall cost of energy is required before a study of the optimal routes forward for commercialisation of the sector can be presented.

Predictions for future LCOE assume that there is a learning rate associated with increasing device deployment or, in other words, the cost of producing the first unit is likely to be much higher than producing the 100th unit. MacGillivray et al. [7] studied the learning rates of ocean renewables, investigating the investment required to bring wave or tidal energy to a level that is cost competitive with offshore wind. The results of the paper imply that the ocean energy sector must achieve cost reduction earlier and with a lower level of cumulative deployment in order to be cost-competitive. MacGillivray suggest that this would be manifested through rapid design consensus and greater levels of shared learning within the sector.

Importantly learning related cost reductions will not be realised without deployment of initial small arrays. Certain ocean energy industry based reports estimate the costs for 20 MW arrays and project them forward [8], but there is currently no literature investigating the costs of first-deployment converters. Final investment decisions on first deployments requires a greater understanding of the current costs by stakeholders capable of impacting ocean energy development.

The purpose of this paper is to ascertain the factors that have the greatest impact on the LCOE of ocean energy technology, and to define how different geographic regions and specific technology optimisation can impact the overall attractiveness of wave and tidal energy economics. A comprehensive PESTLE (Political, Economic, Social, Technological, Legal and Environmental) analysis is proposed in order to reveal the risks and uncertainties that wave and tidal energy technology faces within each of the identified headings. The final outcome of the paper is to present sensible pathways towards sector development that could enhance cost effective energy production, giving greater clarity on where investment and research and development effort should be focused.

This paper is structured into 5 sections, including the introduction. Section 'Methodology' presents the methodology that will be followed in this analysis, outlining the process used to establish the most appropriate factors and case studies. Section 'Deployment case study (geographic)' presents a "deployment" case study, seeking to identify the geographical implications of wave and tidal energy deployment across 6 selected locations. These locations will then be assessed in terms of deployment attractiveness. Section 'Development case study (technology)' presents a "development" case study, exploring the technical factors that affect the LCOE of specific devices, and investigates the effects of a number of scenarios on the overall LCOE for the different technologies under consideration. Finally Section 'Conclusions' concludes the findings of this study.

Methodology

The original aim of the paper was to investigate the factors that affect the LCOE for ocean energy technologies. In order to comprehensively review the risks, uncertainties, and plausible impacts on LCOE a PESTLE analysis was proposed. PESTLE analysis is a common tool used in a number of industries to assess the factors which influences cost [9]. The PESTLE headings represent the following aspects: Political, economic, sociological, technological, legal, and environmental. Within each heading, a number of possible influencing factors were investigated. The factors represent areas in which significant effort must be made to reduce risk and uncertainty within ocean energy project development with varying levels of urgency.

The analysis carried out within this paper identified 93 factors that could impact the development of the wave and tidal energy sectors. Each factor was then scored using a number of metrics, outlined in Table 1 below, providing a ranking of relative impact on LCOE and the progression of the wave and tidal energy sectors

While the PESTLE analysis is exposed to subjectivity, it is not intended for one-off use and should be utilised regularly to ensure that changes in factors are identified and exploited. It is therefore recommended that this work be repeated in the future to ascertain the level of any future developments and sector progression that may take place.

The metrics above have been chosen based on industry research at both Member State and European level [10,11]. Although the initial proposal was to directly investigate, score, and rank the identified factors in order to assess their overall importance on the LCOE of ocean energy technologies, the results clearly showed that there were two distinct themes within the identified factors; a differentiation that merited further investigation.

While some factors could clearly be linked with measurable increases or decreases in LCOE, reliability for example, other factors are difficult to quantify in terms of cost, such as public favour for ocean energy. The factors which directly impacted LCOE were device specific and fell primarily under the "technological" PESTLE heading, with certain "environmental" factors related to resource also having direct impact on LCOE. This was unsurprising, given that LCOE is calculated using CAPEX and OPEX costs (technology related), and using the AEP (which is related to both technological and environmental/resource factors).

By just considering factors which affected LCOE, factors that had the potential to radically alter the route to deployment and/or commercialisation could have been lost. There was a need therefore to group the factors into those which had a direct impact on cost and to group the significant number of factors that did not have a directly quantifiable cost impact but which affected other important parameters such as project development timescales.

In order to be able to assess the importance of each factor in a manner that included technical and non-technical factors and also both device-specific and non-device-specific factors, a new line of approach was required. A decision was made to carry out two separate case studies and provide further granularity to the scoring and ranking of the PESTLE factors. The two case studies were as follows:

1. Deployment – the non-device-specific factors of wave and tidal energy, providing an indicator of attractiveness for deployment of initial ocean energy arrays from a geographic perspective.

Table 1
Scoring criteria for assessment of factors identified within the PESTLE analysis.

Metric	Description	1	2	3
Urgency	How urgent is it that this factor is more comprehensively understood?	Low	Medium	High
Past Experience	Can past experience from other sectors aid and improve current knowledge in this area?	Low	Medium	High
Likelihood of Impact	What is the likelihood of this factor directly impacting the LCOE?	Low	Medium	High
Impact on CAPEX	What effect will this factor have on CAPEX costs?	Low	Medium	High
Impact on OPEX	What effect will this factor have on OPEX costs?	Low	Medium	High
Impact on Power Production - Performance	What effect will this factor have on power production values? (Improving power capture)	Low	Medium	High
Impact on Power Production - Reliability	What effect will this factor have on power production values? (Improving reliability/reducing downtime)	Low	Medium	High
Impact on Accessibility	What effect will this factor have on accessibility costs?	Low	Medium	High

2. Development – identifying the device specific factors that are of highest priority for achieving cost reduction and performance improvement from a technological perspective. In addition to technological factors, it was also agreed that certain factors related to the resource would also have implications on the technology (e.g. weather windows for maintenance intervention).

The 93 identified PESTLE factors were divided into their relevant case study. In order to assess their true impact on the development of the wave and tidal energy sectors, the two separate case studies superseded the original PESTLE analysis ranking and an overall attractiveness index was generated in order to collate the results from each case study.

There is a relationship between technology development and technology deployment that requires both work streams to progress in partnership for the ultimate success of the ocean energy sector. This is consistent with the recommendations within recent European ocean energy research activity [12]. These two case studies will be used in order to assess the readiness of technology and the suitability of deployment sites, assessing the urgent focus areas required in order to accelerate levels of ocean energy deployment.

Deployment

Deployment is the physical installation of ocean energy technologies, and relies heavily on the support of a given state or region to facilitate the route forward to reaching deployment targets. By considering political, economic, sociological, non-device-specific technological, legal, and environmental factors this case study aims to identify the countries that offer the greatest potential for early-stage ocean energy deployment and provide guidance on how other countries could improve on their deployment "attractiveness".

Research into the deployment case study revealed a need to separate out two types of factors: Fixed, and variable. "Fixed" factors are geographical and resource related factors that are not controllable, and have no potential for change, for example, the level of practical resource or geophysical sea bed conditions such as water depth. "Variable" factors have the potential for positive or negative change, either by intervention from policy makers or governments, or through the sociological interaction of various ocean stakeholders, such as fishing and transport.

Due to the deployment factors produced within the PESTLE analysis being difficult to quantify numerically a set of questions were generated which provided quantitative indicators as an outcome. As detailed in Table 2, a set of 16 common "variable factor attractiveness indicators" were devised for this study, with a further 4 wave specific "fixed factor attractiveness indicators", and 5 tidal specific "fixed factor attractiveness indicators". These indicators covered the key aspects of the PESTLE analysis and allows data from relevant countries to be collated, assessed and compared.

Within this study the fixed and variable factors were analysed for each technology providing the relevant data for six countries with a significant interest in ocean energy: the USA, UK, France, Canada, Spain and Chile. A weighted scoring matrix was used to evaluate attractiveness for each country and this is presented graphically. Each indicator was scored on a scale of 0–3, based on performance – 0 representing poor performance and 3 representing excellent performance. The scoring matrix is outlined in Appendix A at the end of this paper.

The six countries selected for this study are sufficient for exploring the range of Ocean Energy support available and for testing the PESTLE methodology. However, this is not to suggest that Ocean Energy development is restricted to these six countries. In particular, countries surrounding the Western Pacific have begun to explore Ocean Energy in recent years and have undertaken fundamental research and development. These include Japan, South Korea, China, Singapore, Australia and New Zealand [13].

Most of these countries have completed detailed resource assessment and have begun prototype testing. Japan, Korea, China and Singapore have dedicated open sea test centres planned or under development, including a 5 berth 5 MW capacity wave test centre in Korea, a full scale wave energy site and full scale tidal site in Zhejian and Guandong, China (respectively) and a tidal site between Singapore and Sentosa Island. Meanwhile projects such as the Korean 200 kW OTEC plant built in 2014 continue to be active. The planned pre-commercial scale deployment of three wave energy devices near Perth, Australia, will be a significant development in the sector [14]. These countries offer huge potential for the sector and are an exciting new market for developers; they are only excluded here in order to keep the content succinct.

Development

The second case study considers technological and resource (environmental sub-set) factors that have a directly quantifiable impact on LCOE. In order to reflect the diverse and different needs of wave and tidal energy, the two technologies were considered separately with a different set of scenarios. For the wave technologies a fixed multi-point absorber and a floating multi-body pitching device were considered. For the tidal technologies three options were considered: a fixed bottom mounted single rotor turbine, a fixed surface piercing foundation structure with twin turbine rotors, and a moored buoyant platform with twin turbine rotors.

In tidal energy, the dominant variable is the current velocity, and this has the largest impact on defining how attractive a certain site will be. For this reason, economically attractive tidal stream array projects exist only in discrete geographical locations with high current velocity. The periodic nature of the tidal forces means that current velocities at these locations can be predicted indefinitely if appropriate acoustic measurements are taken. Within the sector design consensus has largely been achieved and most

Table 2
Attractiveness indicators.

	Attractiveness Indicator	Number	Factor
Variable factors	Number of projects under development[a]	1	Political/ Technological
	Realistic Ocean Energy targets for 2020/2030	2	Political
	Security of supply: Does the country require ocean energy to meet renewable energy targets? (Ocean energy 2020 target expressed as a% of renewable energy target)	3	Political
	Carbon saving: How urgent is the requirement for carbon saving? (Current Renewable Energy Supplied expressed as a% of Total Energy Demand)	4	Political
	Examples of market pull support mechanisms	5	Political/ Economic
	Examples of capital grant support mechanisms	6	Political/ Economic
	"Best 3 Years of Available Data" R&D expenditure - Low TRL	7	Economic
	"Best 3 Years of Available Data" R&D expenditure - High TRL	8	Economic
	Examples of indigenous private sector companies taking a significant interest in ocean energy	9	Sociological
	How many companies/technologies at demonstration level are presently active in the country?[a]	10	Sociological
	Relevant industry and commerce for ocean energy sector learning	11	Sociological
	Level of infrastructure readiness: Test facilities[a]	12	Technological
	Level of infrastructure readiness: Grid	13	Technological
	Level of infrastructure readiness: O&M ports and harbours (number of suitable O&M ports within 300 km radius of a reference site)	14	Technological
	Level of infrastructure readiness: Deep water ports and harbours with manufacturing and large-scale component lay down capability (available, under construction, in planning, not available)	15	Technological
	How complex is the permitting and consenting application process – regulation & legislation	16	Legal/ Environmental
Fixed factors (Wave)	Level of practical resource - wave	17	Environmental/ Resource
	Seasonal variation - Wave	18	Environmental/ Resource
	Wave Mean/Wave Std. Dev.	19	Environmental/ Resource
	Weather window - accessibility time: Suitability for maintenance access (average annual time, expressed as%, at reference site)	20	Environmental/ Resource
Fixed factors (Tidal)	Level of practical resource – tidal	17	Environmental/ Resource
	Water Depth	18	Environmental/ Resource
	Maximum Tidal Velocities	19	Environmental/ Resource
	Tidal Range	20	Environmental/ Resource
	Weather window - accessibility time: suitability for maintenance access (average annual time, expressed as%, at reference site)	21	Environmental/ Resource

[a] Separate information collected for wave and tidal energy.

tidal stream technology, to date, has been designed for similar high velocity resource conditions, with the exception of one developer [15].

The wave energy resource, on the other hand, is highly variable – with at least two governing parameters (significant wave height and wave period). This presents a unique design challenge that is dependable on geographic location which has a large effect on the resource. Therefore, the geographic technology attractiveness study considers only wave energy technology.

LCOE depends predominantly on three parameters: Capital costs (CAPEX), operating costs (OPEX), and Annual Energy Production (AEP). The technology case study will assess the impact on the overall LCOE using a number of defined scenarios, determining which factors have the biggest influence on overall LCOE and how research support, development support and investment should be targeted in order to achieve LCOE reduction.

Attractiveness index

The outcomes of these two identified case studies were collated to create an attractiveness index. To quantify development attractiveness the ratio of the LCOE (calculated for each device and each location) to a target 'economically competitive' LCOE (set at 0.25 €/kW h) was calculated and converted to a percentage. The deployment attractiveness was calculated by multiplying the score for the fixed factors by the score for the variable factors and converting to a percentage. These percentages were then combined to provide the overall attractiveness index. This is used to presents both the geographic and technological impact of a range of factors on the future of the wave and tidal energy sectors. The output of this paper will be a graphical representation of the attractiveness for a given technology in a given location, based on fixed, variable, and technology specific factors.

Summary

A flow chart, presented in Fig. 1, reveals the work path that was taken for this study. The initial PESTLE analysis resulted in the formation of two work streams, deployment and development. These two work streams were further developed and enhanced with fixed and variable factors found within the deployment case study, and a suite of scenarios defined for the development case study. The overall findings of each case study and the respective scenarios culminate in the overall attractiveness index.

The final outcome of the paper is to present a sensible pathway towards sector development that could enhance cost effective energy production, giving greater clarity on where investment

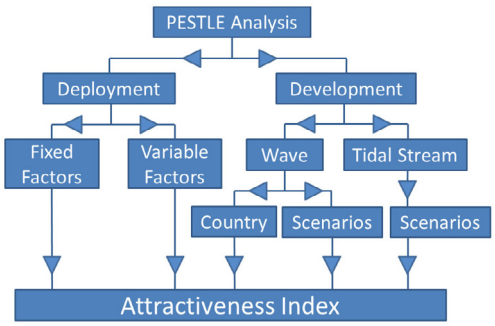

Fig. 1. Methodology flow chart.

should be focussed from both a geographic and a technological perspective.

Deployment case study (geographic)

For this case study, a number of countries were selected which have expressed an interest in developing and deploying ocean energy technology. A set of Attractiveness Indicators were created in order to provide a quantitative assessment for the most important factors that affect the suitability and attractiveness of a country from a deployment perspective.

Firstly, the fixed factors related with the resource (those which cannot be affected by human decisions) are presented for the USA, UK, France, Canada, Spain and Chile based on specific sites (mainly established test sites). In UK, France, and Spain three established test sites have been selected for comparison for wave energy (EMEC, SEM REV, and BIMEP respectively). For the remaining countries, sites with prospective interest on wave energy development and wave energy data have been selected (Humbold Bay for the USA, Nova Scotia in Canada and offshore of Santiago de Chile). For tidal energy, established test centres have been used where appropriate, such as EMEC (UK) and Fundy Ocean Research Centre for Energy (FORCE) (Canada). Prospective deployment sites for tidal energy have been used for appropriate data related to tidal energy projects (e.g. Paimpol Brehat (France), Pudget Sound and Cobscook Bay (USA), Chacao channel and Primera Angostura (Chile) where no test facility currently exists.

Secondly, the variable indicators (those subjected to political and/or economic decisions) are presented in Table 5. Data for these indicators has been taken from publicly available sources and presented here for quantitative comparison.

It should be noted that the indicators identified in Table 5 are all changeable, i.e. they can be improved (or can deteriorate), and the indicators under investigation here are subjected to the influence of political will. Thus the influence of a given factor on geographic attractiveness could change in the future.

Based on the indicators identified in Tables 3–5, the countries were assigned scores according to the scoring matrix (appendix – Appendix A). A single axis comparison of attractiveness would not represent the true region deployment attractiveness, as certain factors can be influenced whil other factors cannot be altered. For example, economic support for ocean energy, in terms of technology-push and market-pull support mechanisms, can be influenced by a government, and as such are susceptible to change (for better or for worse). Factors such as the number of consented projects within a given region will also change as the sector progresses. Infrastructure readiness, in both grid connection capability and ports and harbour facilities, has the potential to create significant change in the level of attractiveness for a given geographic location. The available resource, bathymetry characteristics, tidal range and weather windows are examples of fixed constraints that cannot be influenced through policy measures.

The results of this case study are presented on a wave energy two-axis graph and a tidal energy two-axis graph (Figs. 2 and 3 respectively). In both cases the scoring of natural environmental and resource factors (Fixed factors) can be found on the y-axis, and the scoring of techno-economic and political factors (Variable factors) can be found on the x-axis. The scale is such that the maximum of 1 represents a score of 100% within the relevant set of attractiveness indicators. The actual score achieved by each country will be a fraction between 0 and 1, with higher scores reflecting higher overall attractiveness. A perfect score would result in a country being placed at the coordinate (1, 1). The overall deployment attractiveness is the distance from the origin or (the multiple of the fixed-factor score and the variable-factor score) converted to a percentage.

By presenting the attractiveness in a two-axis graph, a comparison can quickly be drawn between each country based on fixed attractiveness indicators, variable attractiveness indicators, or both simultaneously. This method also highlights the ability for a country to change its current attractiveness for deployment (alter its x-axis score).

Wave

In Fig. 2 the fixed and variable attractiveness indicators for wave energy are plotted. From a variable factor viewpoint the UK is the most attractive country for wave energy. Within the UK support towards wave energy is generally high; stable political support mechanisms have been in place for a number of years, pushing the advancement of the sector. More recently, confidence in the robustness and stability of this political support has been eroded. For instance, the transition from the Renewable Obligation (RO) scheme to Contracts for Difference (CfD) presents a lack of long term certainty for the ocean energy sector. Nevertheless the UK remains at the head of the pack and has a well-established test facility. In terms of fixed factors, although the wave energy resource is abundant, the availability is relatively low-medium because of a harsh operating environment (with conditions that exceed the operating parameters of devices). The overall scoring on the fixed factors is therefore medium level.

The USA has been supporting wave energy since 2011, and they recently provided the ocean energy sector with some R&D funding calls. The country also has great conditions for ocean energy development: the wave energy resource and availability, mainly on the west coast, resulting in a high attractiveness score

Despite having a relatively high wave resource, France seems to be politically focused on tidal energy deployment. However the

Table 3
Fixed factors wave.

Indicator	A	UK	France	Canada	Spain	Chile
Average practical wave energy resource (country)	1170 TW h/yr [16]	42 TW h/yr [17]	40 TW h/yr [18]	240 TW h/yr [19]	25.4 TW h/yr [20]	433 TW h/yr [21]
Coefficient of variation of wave resource [22]	1.2	1.8	1.6	1.5	1.6	0.7
Accessibility [22]	50%	30%	60%	65%	65%	15%

Table 4
Fixed factors tidal.

Indicator	USA	UK	France	Canada	Spain	Chile
Level of practical tidal resource (TW h/yr)	~20 TW h/yr [23]	20.6 TW h/yr [24]	14 TW h/yr [18]	~36.5 TW h/yr [19]	1.84 TW h/yr [25]	4 TW h/yr [21]
Water depth range (m)	26–58 [26,27]	16–116 [24]	30–45 [28]	~53 [29]	~30 [25]	~40–70 [21]
Maximum Current Velocity (m/s)	>3 m/s [27]	>3 m/s [30]	>3 m/s [31,32]	>3 m/s [33]	1.5–2 m/s [25]	>3 m/s [21]
Tidal Range	Up to 7 m [34]	Up to 4 m [30]	Up to 11.8 m [31]	Average 14.5 m [33]	Up to 4 m [25]	Up to 7 m [21]
Site wave sheltering (Full shelter/Partial shelter/Exposed)[a]	Full sheltered sites	Full sheltered, partial sheltered, and exposed sites	Partial sheltered, and exposed sites	Partial sheltered sites	Partial sheltered, and exposed sites	Full sheltered sites

[a] Based on author observation of the land mass shielding around viable tidal-stream energy sites, and principal wave directions.

resource and other factors maintain an established high attractiveness ratio even for wave energy.

Canada has a good level of wave energy resource, with accessibility among the highest at 65% of the year, however as it can be seen on the figure Canada's government is focused on promoting the tidal sector and the support towards the wave sector is still uncertain.

On the other hand Spain has great conditions for wave energy deployment on the north coast but, due to the financial recession, the support towards ocean energy has been very modest during the last 2 years along with the feed-in tariffs for renewables in general [35].

Chile has very high levels of resource, but this is consistently strong throughout the year, leading to accessibility issues, which could be problematic for installation and maintenance. In addition, support mechanisms are not yet in place, and a very weak regulatory system prevents Chile from progressing up the rankings. Consequently, this means the country has a great attractiveness potential for deployment and with the adequate politics/financial changes it could be a future leader within the sector.

Tidal

In Fig. 3 the fixed and variable attractiveness indicators for tidal energy are plotted. In the tidal energy case study the UK is, at present, recognised as the most attractive country for development in terms of overall attractiveness indicators with significant score on both the fixed and variable factors. However, there are limitations, the UK is penalised for having poor grid connection availability in areas of high resource, and certain sites in the UK are very exposed to the action of waves. A well-established test facility, together with a number of consented tidal energy projects, gives the UK significant justification for the leading role in terms of attractiveness for project development. This is reflected in the advanced stage of prototype testing that has already taken place within the UK. Strong levels of government support have been backed up by both capital grant and market pull support mechanisms.

However, as described in the wave analysis, the long-term support mechanisms for tidal energy projects are not as certain as they have been in the past. This situation will need to be rectified in order for the UK to retain its leading position, as the attractiveness margin between the UK, France and Canada is very small. As countries outside of the UK can accelerate the development of the wider sector and supply chain, it may be that these countries will soon level or overtake the UK in tidal energy deployment attractiveness. The leading position held by the UK is primarily now due to legacy effects of having started early with tidal energy support; the country was the first to establish a "one-stop-shop" consenting process, streamlining the consenting of ocean energy projects – requiring project developers to submit an application to one centralised organisation, instead of several independent consultees.

With only a slightly lower theoretical resource than the UK, France follows in a close second in the attractiveness index. France already has good grid connection opportunity and port and harbour infrastructure close to deployment locations, thus scores strongly in the variable factor axis. With strong government support through capital grants and feed in tariff, appropriate market mechanisms are in place. The resource potential is good, although somewhat lower than the resource potential within the UK, USA and Canada. A more complex planning regime detracts from the high scoring. It should be noted that the north of France, where the early array projects will be deployed, does not suffer from the same grid restraints as many of the UK wave and tidal energy sites, potentially fast tracking some of the early demonstration projects, if funding mechanisms provide the correct balance to allow projects to reach a positive final investment decision.

Canada achieves third position with a favourable high resource but there are increased challenges associated with the deep water sites and increased tidal range experienced in the Bay of Fundy's tidal stream sites. However, some of the maximum velocities experienced in the primary tidal stream locations are higher than those in the attractive deployment sites in the UK and France. In the short-term, this will prove to be an engineering challenge that could add additional costs to foundation and mooring structures, although if overcome could allow increased revenue from enhanced electricity generation. The Canadian tidal stream sites are largely sheltered from the worst storm waves.

The USA has good levels of resource, and sheltered sites at favourable water depths for early array deployment, scoring highly on the fixed factors. However, the support mechanisms and political drive for tidal energy are behind that of other countries, so the USA lags behind in the variable factor axis.

Spain, although scoring highly in terms of site accessibility and wider infrastructure, offers only a modest tidal energy resource. With lack of support for feed-in-tariffs, the economic case for tidal energy deployment in Spain is limited. Overall, Spain only scores modestly in both fixed and variable fields: The fixed environmental factors due to only having access to a very modest tidal resource, with relatively low spring peak tidal velocities; the variable factors due to a lack of concentrated effort in supporting emerging renewable energy technologies with feed-in-tariff mechanisms.

Chile has a modest overall technical resource, but has the benefit of strong current velocities in the geographic regions where tidal flows exist. The tidal sites are sheltered from the impact of storm waves. However, there is very little infrastructure to support the development of tidal energy in Chile, and there are no economic support mechanisms in place or formal ocean energy targets. The level of political will is, at present, low. However, with the correct changes implemented, Chile could gain significantly in attractiveness, as the low score is predominantly due to poor performance in the variable factors.

Table 5
Attractiveness indicators.

#	Indicator	USA	UK	France	Canada	Spain	Chile
1a	Number of projects under development (Wave)	10	8	5 [35]	5 [36]	5 [37]	1
1b	Number of projects under development (Tidal)	3	5	4	3	3 [37]	1
2	Realistic Ocean Energy targets for 2020/2030	15 GW by 2030 [38]	300 MW by 2020/4 GW by 2030 [39]	380 MW [40]	250 MW by 2020/2000 MW by 2030 [19]	100 MW by 2020 [19]	NA
3	Security of supply: Does the country require ocean energy to meet renewable energy targets? (Ocean energy 2020 target expressed as a% of renewable energy target)	3.7%	2.2%	0.7%	1.4%	1.3%	NA
4	Carbon saving: How urgent is the requirement for carbon saving? (Current Renewable Energy Supplied expressed as a% of Total Energy Demand)	17% [41]	11.3%	11.6% [42]	65% [43]	15.1% [44]	4.82%
5	Examples of market pull support mechanisms	No confirmed feed in tariff	372 Eur/MW confirmed through EMR CfD until 2019	150 Eur/MW	394.4 Eur/MW [45]	86 Eur/MW [46]	No confirmed feed in tariff
6	Examples of capital grant support mechanisms	NA	WATERS, MEAD MRCF [47]	€80 m (ADEME call for expressions of interest in pilot tidal energy plants)	NA	MINECO: Innpacto 2011, 12,13; Innpronta, RETOS 2014, Cenits CDTI: EEA Grants, Innterconecta [37]	NA
7	"Best 3 Years of Available Data" R&D expenditure - Low TRL (€Million)	24.96 [48]	17.16 [48]	26 [49]	32.4	14 [37]	2.21 [50]
8	"Best 3 years of available data" R&D expenditure - High TRL (€Million)	10.998 [48]	45.16 [48]	0	2.67 [51]	8.5 [37]	0
9	Examples of indigenous private sector companies taking a significant interest in ocean energy	Yes	Yes	Yes	Yes	Limited	Limited
10a	How many companies/technologies at demonstration level are presently active in the country? (Wave)	6	7	0	0	3	0
10b	How many companies/technologies at demonstration level are presently active in the country? (Tidal)	2	10	3	1	1	0
11	Relevant industry and commerce for ocean energy learning [52]	293	52	42	10	8	0
12a	Level of infrastructure readiness: Test facilities (Wave)	4 [53]	3 [53]	1	1 [19]	2 [54]	0 [55]
12b	Level of infrastructure readiness: Test facilities (Tidal)	0	1	1	1	0	0
13	Level of infrastructure readiness: Grid	Grid expansion requirements remain to be identified and defined	Grid upgrades critical to industry development [47]	Future grid upgrades required, however strong capacity in region of first deployments [47]	Grid expansion requirements remain to be identified and defined [19]	Strong capacity in region of first deployments	Grid upgrades critical to industry development [21]
14	Level of infrastructure readiness: O&M ports and harbours (number of suitable O&M ports within 300 km radius of a reference site - site name in parenthesis) [56]	2 (NNMREC); 3 (HINMREC); 6 (SNMREC)	13 (MeyGen); 4 (WaveHub)	8 (SEMREV; 5 (SEENEOH); 6 (Paimpol Brehat)	4 (FORCE)	4 (BiMEP)	2 (Chacao Channel); 1 (Apaio Channel)
15	Level of infrastructure readiness: Renewable energy specific deep water ports and harbours with manufacturing and large-scale component lay down capability	Under Construction: New Bedford Marine Commerce Terminal	Available: Belfast Under Construction: Green Port Hull; Ardersier; Kishorn	Under Construction: Port of Cherbourg	In Planning	In Planning	In Planning
16	How complex is the permitting and consenting application process – regulation & legislation	Multi-agency	Streamlined single-agency	Multi-agency	Multi-agency	Multi-agency	No official process

Development case study (technology)

As specified in the methodology section, the majority of the quantifiable factors that directly affect LCOE are of a technological nature and are directly linked to specific technologies and devices. As a result, this case study was designed in order to address cost and performance elements of technology development. Again, a differentiation between wave and tidal technologies is needed to address the different stages of development of each technology type. The technology case study will analyse specific variants of each technology: fixed and floating.

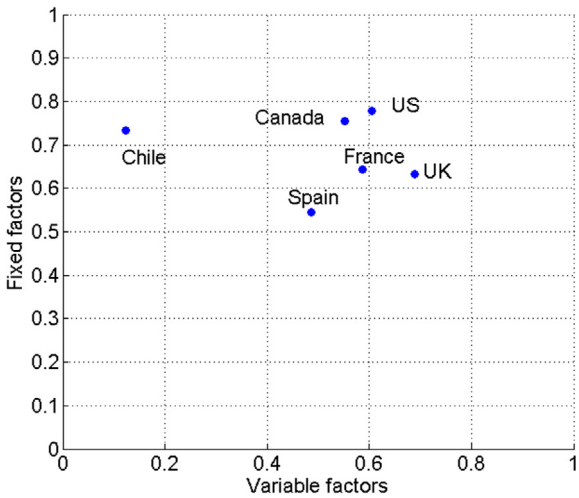

Fig. 2. Fixed vs Variable factors for wave energy deployment.

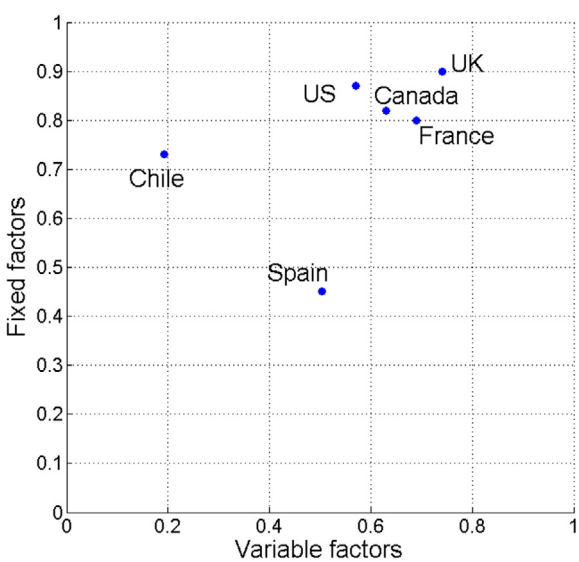

Fig. 3. Fixed vs Variable factors for tidal energy deployment.

Wave technologies

For the wave energy technologies a fixed multi point absorber and a floating multi-body pitching device have been selected for comparison (Fig. 4). The data regarding the energy yield of the devices is given in Table 6. The AEP is calculated by the multiplication of the wave energy scatter plot and the device specific power matrix. The scatter plot has been obtained from a COE Tool from Julia Fernandez Chozas Consulting Engineer [46], and has been applied for both technologies. For the multi-point absorber, the power matrix was obtained through conversation with the developer; for the attenuator, the power matrix was obtained from a desktop study of existing research literature [57]. It should be noted that these figures take into consideration only one unit, and that the learning curve has not been applied to these costs. In Table 7 the Annual Energy Yield of each variation is shown.

In this case study the effect of different variations of the current prototypes on LCOE was investigated, and ultimately, what LCOE could be realistically expected for device deployment in the near future. To achieve this, several variations of each device have been studied. For the multipoint absorber, the base case study has been selected with the following characteristics:

- Structure made of fibre glass.
- Resistive control.
- Hydraulic efficiency of PTO = 0.7.

For the attenuator the base case is set with the following characteristics:

- Reactive control.
- Structure made of steel.
- O&M strategy without optimisation.
- Restrained availability (80%).

It should be noted that in Table 7, each case takes the previous improvements specified in the other cases into account, in order to find out the net influence on the LCOE. For the calculation of the LCOE a discount rate of 15% was applied due to the novelty of the technology. The procedure for cash-flow analysis has been performed following the methodology outlined in previous studies [5].

In Table 8 the LCOE for both prototypes is given. For the multipoint absorber, the base case is very expensive (5 €/kW h) because the control strategy that is applied is not efficient and the prototype is made from glass fibre reinforced plastic (which is relatively expensive). For the second case the control strategy is changed to a reactive control (2 constants, damping and stiffness control strategy), instead of resistive (1 constant control strategy), and, as can be seen in Table 7, the power production sees an increase of almost a 50% (the LCOE is reduced by almost 50%). It also needs to be highlighted that, for the multipoint absorber, the improvement of 10% on the hydraulic efficiency of the PTO leads to a LCOE reduction of 50%.

Also, for both cases the LCOE reduction is very significant when cheaper materials are used for the main hull of the converter. For the multipoint absorber case, ultra-high performance concrete is utilised and leads to a LCOE decrease of 45%. For the attenuator case, similar material improvements and cost reductions lead to a LCOE decrease of approximately 30%.

For the attenuator, the improvement with a highest impact on LCOE is the optimisation of the control strategy of the converter. It is concluded that both converters need to improve their power capture in order to enhance their power production and facilitate an advance towards commercialisation.

However, in terms of comparing the two devices, this comparison is unfair as the deployment locations for both devices are very different (the locations for the multipoint absorber has a resource of just 5.4 kW/m and the location for the attenuator has a resource of 28.5 kW/m). In order to address this, and to reflect the deployment case study, a location is selected within each country. The selected locations are specified in Table 9.

The data for EMEC, BIMEP, Humbold Bay and SEM REV have been obtained from COE Tool from Chozas [46]. The data for Nova Scotia has been obtained from a buoy located at the coordinates (43.71, 59.85) named "wel429", located on a site with depth of 37 m. For the location in Chile, as there wasn't open source buoy data available, the data was selected from a Global reanalysis database used previously in studies by de Andres et al. [22] and validated within existing literature [58], at the coordinates −73, 5, 26.

For these comparisons at a national level, the cases 5M and 5A are selected. The costs are kept the same for all the locations (although it is known that the materials cost and labour cost can vary from country to country). It should be highlighted that no feed in tariff has been applied on this study in order to make a fair comparison among the economic performance among the different

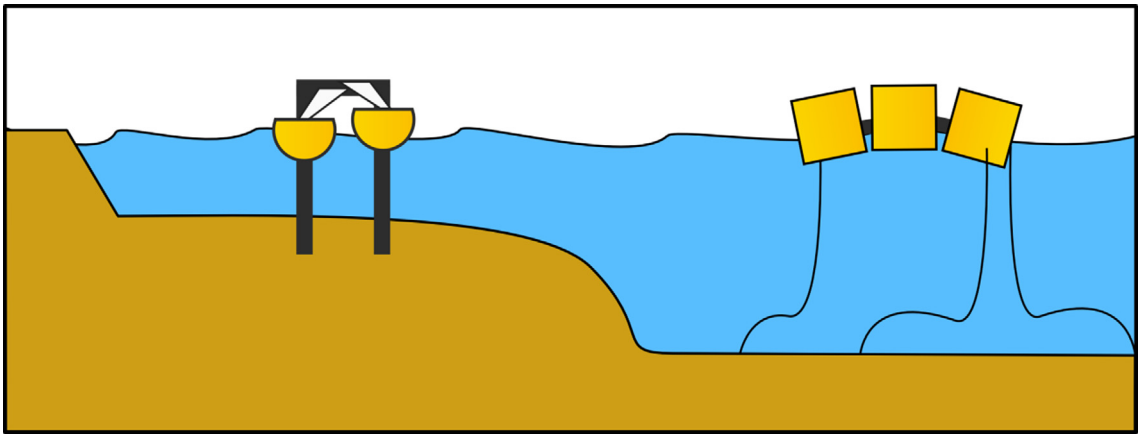

Fig. 4. Wave energy converter technology variations: (from left) fixed multi point absorber and floating multi-body pitching device.

Table 6
Characteristics of the selected devices.

	Multi point absorber	Attenuator
Location	Current Demonstration location for each WEC	
Average depth	30 m	75 m
Average Resource	5,8 kW/m	28,5 kW/m
Estimated lifetime	20 years	20 years
Rated power of the device	1 MW	750 kW
Initial Cost (Structural cost of the device + PTO)	11 M€	3,8 M€
Mooring system	–––––	0,4 M€
Electrical connection	0,3 M€	0,5 M€
Installation	3,5 M€	1,7 M€
Total CAPEX	17 M€	8,3 M€
Annual OPEX	(0,5%CAPEX) 0,18 M€	(3% CAPEX) 0,18 M€

Table 7
WEC characteristics for case studies.

	Scenario	Characteristics	Energy Yield (MW h/year)
Multipoint absorber	1M	Default	340
	2M	Base case+ Reactive control	500
	3M	Case 2+ 10% improvement on PTO efficiency	1028
	4M	Case 3+ Optimum control	1144
	5M	Case 4+ Ultra High perform Concrete	1144
Attenuator	1A	Default	1185
	2A	Base case + concrete	1185
	3A	Case 2 + optimisation of O&M strategy	1185
	4A	Case 3 + enhanced control	1539
	5A	Case 4 + Increase on availability (95%)	1729

locations. The feed in tariff is considered within the political will aspects of the country case study carried out previously.

Fig. 5 shows the LCOE in €c/kW h for each technology and within each location. It should be noticed that for all the cases except EMEC, the multipoint absorber has a lower LCOE than the attenuator. The highest differences are found on Nova Scotia and SEM-REV locations. The multipoint absorber is designed for relatively mild climates and for low energy periods. On the other hand the attenuator is designed for rougher climates and higher energy periods. Then, as this graph shows, both converters have different

Table 8
Affection of improvements on Table 7 on the LCOE.

Scenario	Multi-point absorber	Scenario	Attenuator
Default	5.2 €/kW h	Default	1.4 €/kW h
Reactive control	4.3 €/kW h	Concrete	1.15 €/kW h
+10% PTO efficiency	2.8 €/kW h	Optimized O&M	1.2 €/kW h
Optimum control	4 €/kW h	Enhanced control	1.05 €/kW h
Ultra-high perform concrete	2.9 €/kW h	Increase on availability up to 95%	1.15 €/kW h
Best Case	0.8 €/kW h	Best case	0.6 €/kW h

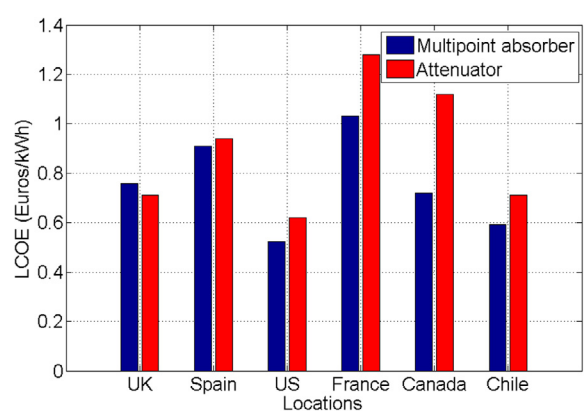

Fig. 5. Cost of energy on the different selected locations.

markets and, as such, their design is different and their prospective target markets should not collide or interfere.

From the power performance point of view it should be highlighted that, as shown in Table 9, the location with highest resource was Chile and the locations with the lowest resource was SEM REV (France). However, on the LCOE chart the location with the lowest LCOE is Humbold Bay in USA. This is due to the fact that Chile, despite its great resource, has a lower availability and therefore net power yield over the period of one year.

Within Nova Scotia the resource is low (16.3 kW/m) compared to the other locations, however, for the multipoint absorber case the LCOE is the third lowest. This leads to the idea that locations with low resource can still be successful for waver energy development if the converter is matched for these particular met-ocean conditions.

Table 9
Selected locations for study.

Locations	Average resource (kW/m)	Availability (%) for multipoint absorber case (Hs < 4 m)	Availability (%) for attenuator case (Hs < 6 m)
United Kingdom (Test location: EMEC)	28.5	92	94%
Spain (Test location: BIMEP)	20.5	99	100%
USA (Test location: Humbold bay)	26.1	95	98%
France (Test location: SEM REV)	15.7	95	99%
Canada (Test location: Nova Scotia)	16.3	99	100%
Chile	30	82	92%

The results shown in Fig. 5 provide evidence to suggest that wave energy converters should be marketed for targeted locations. As the design approach and the working principle is different, the performance of the converters is very different at each of the geographic locations considered within this study. Further study on specific converters is needed in order to gauge the attractiveness of different converter designs for specific wave climates.

The final step was to join the two case studies and calculated the overall attractiveness index. The development attractiveness, described in this section, was calculated as the ratio of the LCOE of each device to a target 'economically competitive' LCOE - set at 0.25 €/kW h, converted to a percentage. The deployment attractiveness, described in Section 'Deployment case study (geographic)', was calculated by multiplying the fixed-factor score by the variable-factor score and converting to a percentage. These percentages were then combined to provide an overall attractiveness index for each country and each wave energy technology examined and this is presented graphically in Fig. 6.

It is important to note that although the deployment case study was made on a country basis, the development case study was much more site specific and dependent on the locations selected for this study.

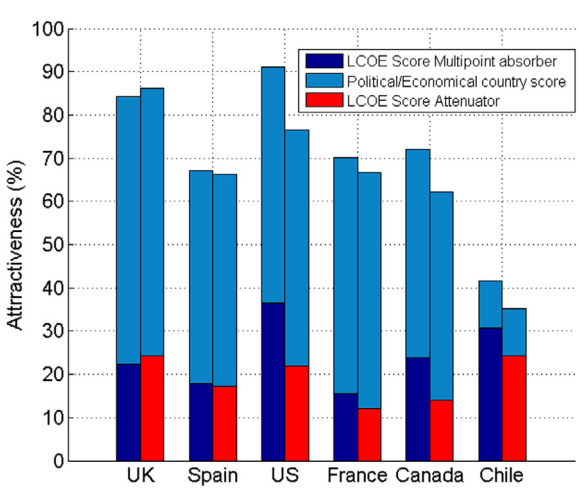

Fig. 6. Attractiveness of the studied technologies at different locations.

The most attractive technology/location combination is the multipoint absorber in the USA. The development case study showed that this location had a low LCOE and the deployment case study showed that the USA overall provided relatively strong political support as well as market-pull funding mechanisms.

In the deployment case study the UK seemed to be the most attractive country for wave energy development owing to the strong political support as well as the financial-push mechanisms. However, in Fig. 5 the attractiveness ratio was not as high as expected. This demonstrates that although the political, economic and social aspects have great importance, the technological barriers are key in order to attract investors.

The opposite is true for Chile. In the deployment case study Chile scored low due to the lack of readiness of the infrastructure within the country, and late emergence of support for ocean energy. However, the LCOE for the two analysed technologies was low, due to the good resource of the Chilean coast. Overall the attractiveness ratio of each technology at the Chile location is therefore moderate, balancing the low social, political and economic attractiveness with the high score for LCOE.

In Canada, the deployment case study showed that the political, economic and social attractiveness for ocean energy is medium and is improving towards a high attractiveness. In the case of the multipoint absorber the LCOE was low and therefore the overall attractiveness of this technology in Canada is high.

Tidal technologies

This section is comprised of work that was also presented in [59]. Three types of tidal energy device were selected for comparison in this case study (shown in Fig. 7). Due to the general convergence of tidal energy converters to horizontal axis turbine designs the technology selected for this study varies primarily in terms of foundation type: a fixed bottom mounted structure with single turbine, a fixed surface piercing structure with twin turbines, and a moored buoyant platform with twin turbines.

For economic project performance a mean spring peak current velocity (v_{msp}) in excess of 2.5 m/s is desirable. While tidal stream resource exists in a variety of locations globally, the same principal flow characteristics are required in order to identify desirable hotspots, regardless of the geographic location. The required data and resolution for a range of locations is not available at this time and therefore this study does not include geographic variation in its assessment of attractiveness for tidal stream technologies. Instead, a representative site was used for all technology variations. Access to high fidelity acoustic current measurements allowed the creation of a velocity profile throughout the entirety of the water column, depth of a device being the primary variation in this study.

The two fixed technologies are assumed to operate lower in the water column than the moored buoyant device. As a result of velocity shear, the velocity profile within a tidal channel generally follows an exponential profile; 75% of the energy in a tidal stream is located in the upper 50% of the water column [60]. The moored buoyant device is assumed to have access to faster flowing water closer to the surface, giving it enhanced AEP characteristics in comparison to the fixed turbine variants.

Utilising information acquired during the Strategic Initiative for Ocean Energy (SI Ocean) project [8], cost estimations and approximate cost breakdowns for current generation tidal energy technologies have been combined with the Carbon Trust's Cost Estimation Methodology tool [61] in order to assess the LCOE resulting from these devices. It must be noted that these cost estimates are for current generation technologies (i.e. the devices that will be used within the initial, pre-commercial, array projects), and therefore there has been no assumed cost reduction due to learning effects.

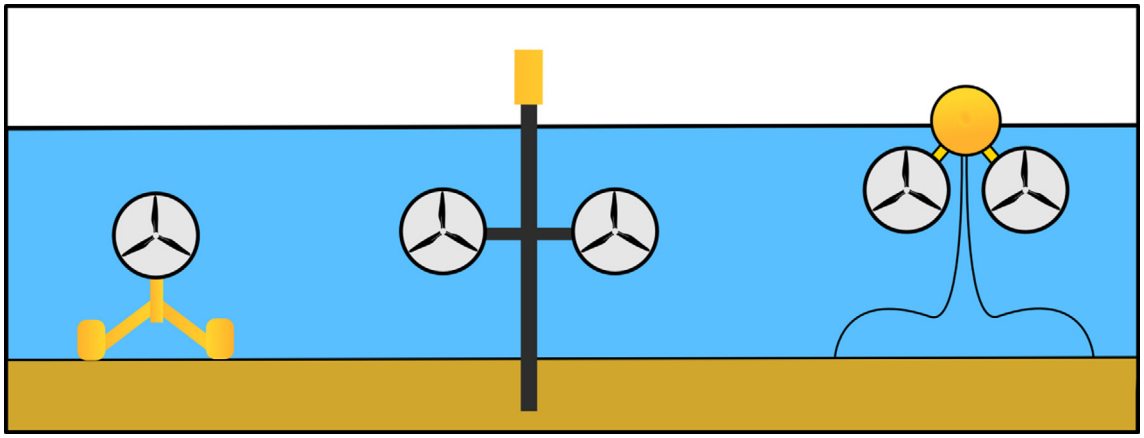

Fig. 7. Tidal energy converter technology variations: (from left) fixed bottom mounted, fixed surface piercing, moored buoyant.

In order to be able to compare the technologies, consideration must be given to existing cost structures, as the current costs of tidal technologies will play a significant role in the route to project development and deployment. There is little benefit in using assumed future costs if the current costs of tidal energy projects are preventing early arrays from reaching Final Investment Decision (FID).

The base case scenarios for each foundation type are based on information available in the public domain, with validation through the SI Ocean project for cost breakdown percentages by cost centre [8] providing a representation of costs with generic turbine Power Take-Off (PTO) and electrical conditioning components. The installation and O&M costs are foundation specific. A basic summary of the input values is shown in Table 10 below. Certain information is confidential (e.g. annual insurance costs, annual OPEX costs and the lease rates associated with Crown Estate seabed rent), and has not been replicated within this report, but an indicator of the relative cost values has been provided.

Base case assumptions

For the purpose of this paper, it has been assumed that each rotor consists of a 1 MW powertrain, with identical performance characteristics for each rotor, regardless of foundation type used. Therefore the fixed bottom mounted device is assumed to have a 1 MW rated capacity, while the fixed surface piercing and the moored buoyant platform options are assumed to have 2 MW rated capacities (due to the twin rotors in these designs).

Using a reference current velocity dataset (collected using acoustic measurement of the current velocity), consideration was made for the approximate hub height for the bottom mounted designs – set at a value of 25 m above the sea bed – and moored buoyant designs – set at a value of 35 m above the sea bed (taken as an approximation of 12 m below the mean sea level for simplicity, however in reality it is recognised that the height above the sea bed will vary constantly with a moored buoyant turbine design). AEP calculation was based on a generic power curve. Current velocities recorded using an Acoustic Doppler Current Profiler (ADCP), over a 31 day time period at a 1 min sample interval, was used as the input data, and then harmonic analysis (further details of this process can be found in literature [62,63]) was performed in order to recreate a velocity time series for a one year period. This one year dataset of velocities was used to calculate an approximate AEP at both 25 m and 35 m hub heights.

Table 10
Tidal technology assumptions.

	Fixed Bottom Mounted	Fixed Surface Piercing	Moored buoyant
Device Rated Power (MW)	1	2	2
Annual Electricity Production per device (GW h/year)	~2.6	~5.2	~5.8
CAPEX Costs (€million/MW)	8.6	5.9	3.75
Installation Cost	High	High	Low
OPEX Costs	High	Low	Low
Availability	85%	85%	85%
Design Life (years)	25	25	20

CAPEX costs are linked to installation costs, and the costs associated with the technical complexity of the foundation design and necessary installation technique. This also accounts for the type of vessel required to complete the installation process. Within the CAPEX cost figures, all costs associated with turbine PTO, turbine structure, control and instrumentation, and grid connection are assumed to be identical for each turbine design, regardless of foundation type. However, mooring, foundation and installation costs are assumed to vary depending on foundation or mooring type, predominantly due to the different vessel requirements and quantities of material involved.

CAPEX and OPEX costs for a sea bed mounted turbine are high, due to the need for heavy lift capabilities in order to deploy or retrieve the nacelle to/from the bottom mounted foundation. A return to base O&M technique will be required regardless of the nature or severity of the failure.

For surface piercing bottom mounted designs, there is a similarly high CAPEX cost, but there is an opportunity to carry out maintenance in-situ. As a result of the ability to lift rotors out of the water, vessel requirements for O&M are significantly reduced, favourably impacting the OPEX cost.

For moored buoyant designs, a return to base technique may be needed, although for simple fault correction it may be possible to replace components in-situ. However, given the ease of access to surface mounted platforms, and the reduced vessel requirements, the costs associated with the O&M in-situ, or indeed in a return to base situation, are significantly lower than for bottom mounted designs.

Given that there is no information publicly available on the performance and availability characteristics of tidal devices, an estimated 85% availability has been applied to the base case scenario as an achievable 'target'. Offshore wind, a more mature technology with significantly greater levels of deployment has shown availability levels, in certain cases, of up to 90% [64]. The design life for the bottom mounted designs is 25 years, based on input from the SI Ocean project. The moored buoyant turbine has a 20 year design life. This is based on information obtained from device developers [65].

The baseline inputs, when entered into the appropriate cells of the Carbon Trust Cost Estimation spreadsheet tool [61] provide reference LCOE for each technology type. An assumed discount rate of 15% has been applied leading to the results, in €c/kW h for the base case assumptions, presented in Table 11.

Scenarios

The base case scenario in Table 11 represents the current LCOE of the first tidal energy converters. However, it is not the intention of the sector to remain at these high costs, and significant cost reduction is being pursued in both capital and operating costs. In order to assess the relative impact of small perturbations of CAPEX, OPEX, and AEP on the overall LCOE, a range of scenarios were devised, designed to represent a sensitivity analysis.

In addition, high insurance costs have been cited as a significant cost burden to the tidal energy sector [11], so a scenario investigating the effect of reduced insurance cost (which could be considered to link directly with annual OPEX expenditure) has also been included in this investigation.

The scenarios, and the resultant changes to the LCOE are detailed in Table 12. The best case scenario consists of 50% reduction in CAPEX, 50% reduction in OPEX, 10% increase in availability, and 50% reduction in Insurance costs.

Table 12 above clearly highlights the high LCOE currently experienced by tidal energy. Current LCOE figures are significantly higher than the average EU-28 market cost of electricity (at 20 €c/kW h [66]) and are also higher than the most favourable market pull support mechanisms that are available at present (38.9 €c/kW h in the UK through EMR). The high LCOE outlined above requires support from capital grants or other funding sources, in addition to attractive market pull support mechanisms, in order to make a valid business case to progress with the array deployment. Although the LCOE, at present, appears high, each scenario represents a plausible improvement in LCOE that will help various technology platforms reach a level closer to economic attractiveness. However, it should be noted that the only foundation/mooring solution that offers an LCOE reduction to near-competitiveness with offshore wind (currently ~19.4 €c/kW h [2]) is the moored buoyant platform with twin turbines.

- A 10% reduction in CAPEX resulted in a 6%, 6%, and 5% reduction in LCOE for Fixed Bottom Mounted, Fixed Surface Piercing, and Moored buoyant technologies respectively.
- A 10% reduction in OPEX resulted in a 3%, 2%, and 2% reduction in LCOE for Fixed Bottom Mounted, Fixed Surface Piercing, and Moored buoyant technologies respectively.
- A 10% increase in availability resulted in an 11% reduction in LCOE in all cases.

This sensitivity suggested that modest changes to the AEP resulted in significant contributions to LCOE decrease. Turbine reliability and availability are therefore paramount to achieving a favourable and attractive LCOE. Availability will be impacted by both turbine reliability, and also the weather conditions during which an installation or a retrieval operation is planned to be carried out. The weather, specifically wind speed, significant wave height and period, will affect lifting operations and vessel movements, and this will in turn affect the ability to carry out turbine intervention and O&M activity. The duration of O&M procedures will prove to be a decisive factor in whether a technology can truly achieve a competitive LCOE.

Table 11
Baseline LCOE for tidal using an assumed discount rate of 15%.

Fixed Bottom Mounted	Fixed Surface Piercing	Moored buoyant
91 €c/kW h	61 €c/kW h	45 €c/kW h

Table 12
Scenarios and resultant LCOE in €c/kW h using an assumed 15% discount rate.

Scenario	Fixed Bottom Mounted	Fixed Surface Piercing	Moored buoyant
Default	91 €c/kW h	61 €c/kW h	45 €c/kW h
−10% CAPEX	86 €c/kW h	58 €c/kW h	43 €c/kW h
+10% CAPEX	96 €c/kW h	65 €c/kW h	47 €c/kW h
−50% CAPEX	65 €c/kW h	44 €c/kW h	34 €c/kW h
−10% OPEX	88 €c/kW h	60 €c/kW h	44 €c/kW h
+10% OPEX	93 €c/kW h	63 €c/kW h	46 €c/kW h
−50% OPEX	79 €c/kW h	56 €c/kW h	39 €c/kW h
10% decrease in Availability	103 €c/kW h	70 €c/kW h	51 €c/kW h
10% increase in Availability	81 €c/kW h	55 €c/kW h	40 €c/kW h
−10% insurance	90 €c/kW h	60 €c/kW h	44 €c/kW h
−50% insurance	85 €c/kW h	56 €c/kW h	40 €c/kW h
Best Case	43 €c/kW h	30 €c/kW h	22 €c/kW h

In order to assess more significant cost reductions in both CAPEX and OPEX, reductions of 50% in both cases were also investigated.

- A 50% reduction in CAPEX resulted in a 28%, 28%, and 23% reduction in LCOE for Fixed Bottom Mounted, Fixed Surface Piercing, and Moored buoyant technologies respectively.
- A 50% reduction in OPEX resulted in a 13%, 9%, and 12% reduction in LCOE for Fixed Bottom Mounted, Fixed Surface Piercing, and Moored buoyant technologies respectively.

Very high insurance costs are currently required for this new technology, and these costs have been cited by various technology developers as a significant constraint on the ability of technologies to become cost competitive with other renewable energy technologies. It is recognised that insurance costs are likely to decrease significantly with increasing turbine reliability and performance (and also the involvement of large Original Equipment Manufacturers (OEMs) such as Siemens and Alstom), and so the sensitivity analysis carried out here investigated the impact to the LCOE based on a 10% insurance reduction and a 50% insurance reduction.

- A 10% reduction in insurance costs resulted in a 1%, 2%, and 2% reduction in LCOE for Fixed Bottom Mounted, Fixed Surface Piercing, and Moored buoyant technologies respectively.
- A 50% reduction in insurance costs resulted in a 6%, 9%, and 11% reduction in LCOE for Fixed Bottom Mounted, Fixed Surface Piercing, and Moored buoyant technologies respectively.

In all of the above scenarios, the technology still fails to reach cost competitiveness with offshore wind, and so a combination of scenarios will need to be achieved in order to unlock the large scale deployments that the industry seeks. A best case scenario, with a 50% reduction in CAPEX, 50% reduction in OPEX, 50% reduction in insurance costs, and a 10% increase in availability, was therefore investigated to yield the resulting changes to LCOE.

- A best case scenario resulted in a 52%, 52%, and 51% reduction in LCOE for Fixed Bottom Mounted, Fixed Surface Piercing, and Moored buoyant technologies respectively.

A comparison of the LCOE achieved in the default scenario and the best case scenario for all foundation and mooring types is shown in Fig. 8.

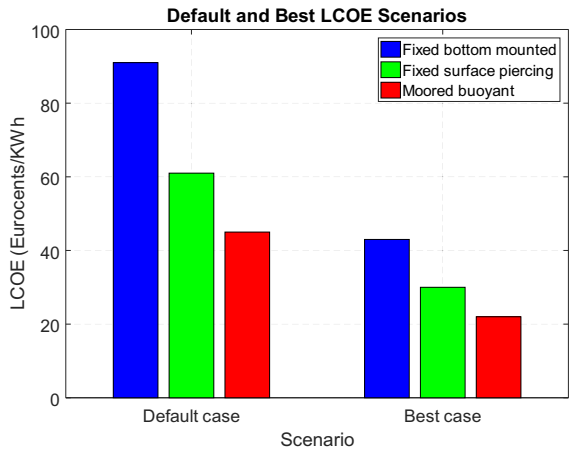

Fig. 8. Tidal - default and best scenario.

It should be noted that the fixed bottom mounted technologies with a single rotor are significantly less attractive in financial terms than twin rotor technology. Even in a best case scenario, a single 1 MW rotor will effectively result in an LCOE 43% higher than that of twin rotors mounted on a surface piercing bottom mounted foundation. It is therefore clear that single rotor turbines at MW scale offer limited opportunity to achieve levels of cost reduction significant enough to enable cost-competitiveness with technologies such as offshore wind. It is unlikely with existing systems and cost structures, that a MW scale single rotor device will be financially competitive with the available alternatives. Multiple rotor structures allow enhanced performance in terms of LCOE, due to the increased swept area allowing increased AEP from a given foundation platform.

However, in all scenarios, the bottom mounted structures are significantly outperformed by the moored buoyant structure. Due to the significant changes to vessel requirements, and the ability to connect and disconnect devices using only modest workboat sized vessels, the moored buoyant technology has, on average, an LCOE of 51% less than the fixed bottom mounted turbine, and 27% less than the fixed surface piercing turbines.

There are significant opportunities to improve the availability and reliability of moored buoyant turbines, without incurring the same intervention cost penalty as would be seen with bottom mounted devices. Significantly greater levels of learning could take place with a much more attractive financial model using surface moored buoyant structures for mounting of multiple turbine rotors.

A range of scenarios are required in order to achieve cost competitive power generation, and the type of foundation or mooring will significantly impact the early array economic case. While significant CAPEX and OPEX cost reductions are essential, modest availability (and AEP) improvements are capable of a more drastic alteration in the LCOE compared to modest CAPEX and OPEX reductions. The relationship between foundation type and device accessibility in the event of a component failure must therefore be given due attention, in addition to the reliability improvements of components and sub-components. Efficient component replacement is essential in achieving high availability in early devices, where initial "bedding-in" failures are more likely to occur.

Conclusions

Ocean energy development and deployment has not yet reached a commercial stage. It is clear that the LCOE for wave and tidal technologies will need to decrease significantly in order to achieve competitiveness with the other renewable sources such as offshore wind. However, within the ocean energy sector, there are factors beyond the LCOE that significantly affect the overall commercialisation trajectory of the wave and tidal energy sectors.

This paper has highlighted that, in addition to the technologically-specific factors, there are several fixed and variable geographical attractiveness indicators that should be considered in order to assess the optimum locations for development and deployment activity.

In wave energy the USA achieves the highest scoring in terms of combined fixed and variable factors. Generally the west coast of the USA has excellent resource levels with a reasonably high availability. The environmental conditions for wave energy deployment are good. Political will is strong, and the government within the USA has been supporting wave energy for a number of years - with a growing R&D investment budget. While other countries show strong resource levels, the variable factors lag behind the USA. The UK leads in the variable factors, due to strong levels of government support, excellent test infrastructure, and attractive feed in tariff levels. However, the resource is more challenging in terms of device availability for power production.

For tidal energy, the UK has a small lead in terms of overall deployment attractiveness. This, however, is primarily due to legacy effects of the UK having started early with tidal energy support, and early establishment of consenting mechanisms and grid-connected testing facilities, in addition to the excellent levels of resource. France, USA, and Canada all show great potential to draw level with, or even overtake the UK as the most attractive location for tidal energy development, particularly if grid constraint issues are removed in these locations in a timelier manner than is being experienced within the UK. Chile, despite significant attractiveness in terms of the "fixed" attractiveness indicators, requires the formation of substantial infrastructure and political frameworks before it can truly establish itself as an attractive location for development – however the potential for improvement is clear.

It has been demonstrated that for wave energy, different technological solutions offer different levels of attractiveness based on the specific resource available in a given geographic location. The technology considered within this paper should therefore not be seen as competitors to one another; actually an adequate market targeting strategy could lead to a range of successful technologies on different locations. While both technologies currently find themselves much earlier in the development life-cycle than tidal energy, there is much potential for the unlocking of significant cost reduction. The greatest reductions in LCOE will come from appropriate material selection, thus reducing CAPEX costs, and optimisation of the control systems to enhance the AEP. Multi-point absorbers have been shown to offer the highest deployment attractiveness within the USA. Attenuator devices are most attractive within the UK.

Tidal energy converters have largely converged around the horizontal axis turbine design, but several foundation and mooring options exist. The dominant factor in LCOE reduction is AEP, therefore improving the reliability and availability, in addition to the maximising capture of the highest energy in the upper parts of the water column, are critical in achieving LCOE reduction. A clear focus of tidal energy converter technology developers must be on achieving and demonstrating reliable turbine performance. Without significant levels of availability, the LCOE quickly increases, and any unplanned interventions could make projects teetering on the edge of viability lose any credibility in achieving significant cost reduction. At this present early stage of development, early bedding in failures and device optimisation may necessitate frequent maintenance interventions, and an accessible platform for turbine retrieval is essential. The analysis undertaken within this research suggests that single rotor turbines located on the sea

bed will not be able to achieve an economically acceptable LCOE to foster continued long term deployment using existing techniques. Multiple rotor platforms enhance the LCOE significantly, particularly if the turbines can be retrieved for maintenance with ease. The most significant cost savings could however be achieved through the selection of moored buoyant platforms, which unlock significant CAPEX and OPEX savings, in addition to an ability access to the higher velocity flows in the upper water column – substantially reducing the LCOE through three key areas in comparison to other foundation options. This work has demonstrated that it is likely that only moored buoyant tidal turbines are capable of cost reductions to below 0.30 €/kW h in the medium term.

While this paper has ranked the attractiveness of several countries, it is recognised that there are additional locations globally where adequate and enticing levels of resource exists. Future application of this methodology on new areas of interest would allow for any new locations to be compared and scored for comparison with the existing data. Indeed, due to the variable nature of many of the attractiveness indicators used in this study, this work should be repeated at regular intervals in order to compare and contrast rates of progress in the different geographic locations under investigation.

Overall, this paper has demonstrated that, while undoubtedly important, there are factors beyond the LCOE that are going to drive the level of deployment activity, and indeed the locations in which this deployment will take place. This paper has presented a realistic assessment of current LCOE costs for a number of present day technologies, and although it is acknowledged that significant cost reduction can be achieved through learning, the technologies that offer the greatest potential for economically attractive deployment may be different than those currently at the forefront of sector development. Focus must be on the AEP from devices, driving improvements in reliability and reducing the length of time taken for intervention operations. If sustainable development and deployment is to be achieved by the wave and tidal energy sector, then the sector must recognise the weaknesses and establish achievable trajectories to overcoming these challenges. This paper has outlined strengths and weaknesses from both a geographic and technological perspective, and has made suggestions on the plausible routes towards a commercially attractive wave and tidal energy sector. The global wave and tidal energy resource is abundant and remains largely untapped, but the opportunities for achieving commercial success are clear. The wave and tidal energy sector must work to achieve the targets that have been set in order to sustain long term credibility and reach the deployment targets envisioned.

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Appendix A

Question		Range	Score
1	Number of projects under development	0	0
		0–5	1
		6–15	2
		>15	3
2	Realistic Ocean Energy targets for 2020/2030	No Target	0
		Unrealistic/ Optimistic Target	1
		Achievable Target	3
3	Security of supply: Does the country require ocean energy to meet renewable energy targets? (Ocean energy target expressed as a% of renewable energy target)	No Information	0
		0–1%	1
		1–3%	2
		>3%	3
4	Carbon saving: How urgent is the requirement for carbon saving? (Renewable Energy Supplied expressed as a% of Total Energy Demand)	0–10%	0
		10–20%	1
		20–50%	2
		>50%	3
5	Examples of market pull support mechanisms	Not Available	0
		<100 €c/kW h	1
		100–300 €c/ kW h	2
		>300 €c/kW h	3
6	Examples of Capital Grant support mechanisms	None	0
		0–10 m €	1
		10–15 m €	2
		>15 m €	3
7	"Best 3 Years of Available Data" R&D expenditure - Low TRL	Not Available	0
		<20 m	1
		20–40 m	2
		>40 m	3
8	"Best 3 Years of Available Data" R&D expenditure - High TRL	Not Available	0
		<20 m	1
		20–40 m	2
		>40 m	3
9	Examples of indigenous private sector companies taking an interest in ocean energy	No	0
		Yes	3
10	How many companies/ technologies at demonstration level are presently active in the country?	0	0
		1–2	1
		2–5	2
		>5	3
11	Relevant industry and commerce for ocean energy sector learning	0	0
		0–20	1
		20–100	2
		>100	3
12	Level of infrastructure readiness: Test facilities	No Test Facilities available	0
		Scale Model Test Tank Facilities	1

Appendix A (*continued*)

Question		Range	Score
		Scale Test Facility (no grid connection)	2
		Full-Scale Test Facility (with grid connection)	3
13	Level of infrastructure readiness: Grid	No Information	0
		Weak	1
		Intermediate	2
		Strong	3
14	Level of infrastructure readiness: O&M ports and harbours (number of suitable O&M ports within 300km radius of a reference site)	0	0
		1	1
		1–3	2
		>3	3
15	Level of infrastructure readiness: Deep water ports and harbours with manufacturing and large-scale component lay down capability (available, under construction, in planning, not available)	Not Available	0
		In Planning	1
		Under Construction	2
		Available	3
16	How complex is the permitting and consenting application process – regulation & legislation	No Information	0
		Multi-Agency	1
		One-Stop-Shop	2
		Streamlined One-Stop-Shop	3

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